

Himasriya Engu

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SUMMARY

AI/ML Engineer, Data Analyst, and Computer Vision Researcher pursuing an M.S. in Computer Science at the University of Central Florida (GPA: 3.92). Published researcher with experience in deep learning, medical image analysis, computer vision, predictive modeling, and exploratory data analysis. Proficient in Python, SQL, Power BI, PyTorch, TensorFlow, and data-driven decision making. Teaching Assistant at UCF with strong foundations in algorithms, data structures, and statistical problem-solving. Seeking internship and research roles in Data Analytics, Machine Learning, Computer Vision, and Medical AI.

TECHNICAL SKILLS

Languages	: Python, SQL, Java, C++, R, JavaScript, MATLAB
AI / ML	: PyTorch, TensorFlow, CNNs, scikit-learn, OpenCV, YOLOv8, Transfer Learning, ResNet, VGG, ViT, Image Segmentation, Data Augmentation, Model Fine-Tuning, XGBoost
Data Analysis	: Pandas, NumPy, Statistical Analysis, Hypothesis Testing, EDA, Feature Engineering, SHAP, Predictive Modeling, Data Cleaning, A/B Testing
Data Visualization	: Power BI, Excel, Matplotlib, Seaborn, Plotly, Streamlit, Jupyter Notebook
Web Technologies	: React, HTML, CSS, Flask, Django, FastAPI, Express.js, REST APIs
Databases	: MySQL, PostgreSQL, MongoDB
Cloud / MLOps	: AWS (EC2, S3), Docker, MLflow, LangChain, ChromaDB, Git, Linux, CI/CD

PROFESSIONAL EXPERIENCE

Teaching Assistant, University of Central Florida, Orlando, FL

Feb 2026 – Present

COP 3502C: Computer Science I

- Support 30+ undergraduate students per week during programming lab sessions and office hours, reinforcing core CS concepts.
- Help students debug code, understand data structures and algorithms, and complete Java programming lab activities.
- Grade assignments for a class of 80+ students, providing structured feedback on code quality, logic, and correctness.
- Collaborate with the instructional team to align grading rubrics and ensure consistent, fair course delivery.

Data Science and Machine Learning Intern, EISystems

May 2024 – Jun 2024

- Conducted exploratory data analysis (EDA) and statistical analysis on 890+ records, identifying key survival predictors through correlation analysis, distribution plots, and feature importance ranking.
- Applied data cleaning, missing value imputation, and feature engineering to prepare structured datasets for machine learning model training.
- Built a Titanic Survival Prediction model using Logistic Regression, Random Forest, and SVM; selected the best model via cross-validation, ROC-AUC, and precision-recall evaluation.
- Visualized model results and data insights using Matplotlib and Seaborn to communicate findings clearly to stakeholders.

Full Stack Web Development Intern, DevTown

Feb 2024 – Apr 2024

- Developed and deployed 3 full-stack web applications using Flask, Django, and Express.js with RESTful API integration and responsive UI design.
- Wrote optimized SQL queries for data retrieval, aggregation, and reporting workflows across MySQL and MongoDB databases.
- Deployed applications on AWS EC2 and implemented JWT-based authentication to improve application security and session management.
- Participated in Agile sprint planning, code reviews, and iterative feature delivery within a cross-functional development team.

PROJECTS & PUBLICATIONS

Eye Deep-Net: Retinal Disease Diagnosis

GitHub — Publication

- Designed and trained CNN architectures with transfer learning for 4-class retinal disease classification (DR, MH, ODR, Normal), achieving ~95% classification accuracy.
- Applied image segmentation and preprocessing including resizing, normalization, contrast enhancement, and data augmentation to improve generalization across imbalanced classes.
- Fine-tuned pre-trained ResNet and VGG models; evaluated with cross-validation, precision, recall, F1-score, and confusion matrix analysis.
- Addressed class imbalance using weighted loss functions and oversampling; benchmarked CNN architectures for optimal model selection.
- Contributed to experimental design, statistical results analysis, and peer-reviewed manuscript preparation and submission.

Tools: Python, PyTorch, TensorFlow, CNNs, OpenCV, ResNet, VGG, scikit-learn Published: TURCOMAT, 15(3), Sep 2024

Alzheimer's Disease Detection using CNN

GitHub

- Engineered a CNN-based Alzheimer's detection pipeline using MRI brain scans with hippocampus-focused image analysis to support automated early-stage diagnosis.
- Compared deep learning architectures across 4 disease stages (non-demented, mild, moderate, severe); selected optimal model

based on sensitivity and specificity tradeoffs.

- Applied MRI preprocessing including skull stripping, intensity normalization, and slice selection; evaluated clinical applicability using accuracy, sensitivity, and AUC.

Tools: Python, CNN, TensorFlow, MRI Imaging, Medical AI, scikit-learn

Real-Time Vehicle Counting and Classification System

Computer Vision, Python, OpenCV, YOLOv8

- Developed a real-time computer vision pipeline for vehicle detection, counting, and multi-class classification from live video feeds using YOLOv8.
- Achieved 90%+ detection accuracy across 5 vehicle classes; processed video at 25+ FPS using OpenCV with optimized inference for varying lighting conditions.
- Evaluated detection performance using mAP and IoU metrics; tuned NMS and confidence thresholds to reduce false positives in high-density traffic scenes.

Tools: Python, YOLOv8, OpenCV, Machine Learning, Computer Vision

Hybrid Movie Recommendation System

Machine Learning, Python

- Combined collaborative filtering (matrix factorization) and content-based filtering to generate personalized recommendations, improving hit rate over baseline.
- Addressed cold-start problem for new users via item content features; evaluated system using precision@k, recall@k, and NDCG metrics.

Tools: Python, scikit-learn, Pandas, Recommendation Systems

EDUCATION

Master of Science in Computer Science

University of Central Florida, Orlando, FL

Aug 2025 – Present

GPA: 3.958 / 4.0

Coursework: Machine Learning, Deep Learning, Computer Vision, Advanced Algorithms

Bachelor of Technology in Computer Science and Engineering

Malla Reddy Engineering College for Women, Hyderabad, India

Aug 2021 – Apr 2025

GPA: 3.56 / 4.0 · First Class with Distinction

Coursework: Data Structures, OS, Networks, OOP Java, DBMS, AI & ML, Data Mining, Big Data

CERTIFICATIONS

- Oracle Certified Associate — Oracle
- Python for Everybody — Coursera
- AWS Cloud Practitioner Essentials — Amazon Web Services
- Power BI Data Analytics Certification — Skill Nation